

Alex Cugnod Segovia

DATA SCIENTIST · STATISTICIAN · MACHINE LEARNING

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PROFILE

Statistician and Data Scientist with hands-on experience building data-driven solutions for industrial manufacturing at Renault Group. Strong background in statistical modeling, machine learning, predictive analytics and data visualization, with end-to-end experience turning fragmented business data into practical analytical applications. Experienced in Python, statistical modeling, time series and Streamlit, with a strong focus on solving open-ended problems independently.

PROFESSIONAL EXPERIENCE

Data Scientist & Software Engineer — Innovation Department | Renault Group | Palencia, Spain | Jan 2026 - Present

Absenteeism Analytics & Prediction Platform

- Built an end-to-end absenteeism analytics platform integrating React, FastAPI, Python, Looker, local datasets and web/JSON sources, working with 3M+ person-day records across 99 variables and growing.
- Developed the application architecture from data processing and API services to an interactive React frontend, replacing fragmented Excel/legacy-tool workflows with a centralized analytical platform.
- Built 10+ functional modules covering factory, department, workshop, UET and employee-level analysis, with cascading filters and interactive data exploration.
- Implemented authentication, sessions, role-based access control, employee management, configuration management and automated reporting through a FastAPI backend and React frontend.
- Developed interactive analytical dashboards with Plotly, including absenteeism evolution, organizational breakdowns, employee tracking, recurrence analysis and operational KPIs.
- Integrated an AI-powered natural-language assistant capable of querying absenteeism data within the user's current organizational scope and returning tables, charts and employee-level results.
- Developed experimental absenteeism forecasting models using Elastic Net, Random Forest and SARIMAX, with calendar features, lagged variables and temporal validation.
- Designed a walk-forward validation framework from scratch to evaluate forecasting performance while avoiding temporal data leakage.
- Worked with managers, HR and operational stakeholders to identify data-quality issues and define consistent absenteeism metrics and business rules.

Production Analytics & Optimization

- Maintained and extended Docuzilla, a Python/DuckDB tool used daily by Renault engineers to identify similar vehicle configurations and investigate production issues.
- Optimized core processing logic, reducing execution time to less than 10% of the original runtime, while developing new features and algorithmic improvements.
- Developed an experimental next-day factory gas-consumption forecasting model using historical consumption, production and meteorological data.
- Designed a PostgreSQL database for quality-management dossiers, including schema, relationships, constraints, indexes and views.

Data Scientist / Analyst | ITACyL — Agricultural Technological Institute of Castilla y León | Valladolid, Spain | Apr 2025 - Jul 2025

- Developed the institute's first predictive modeling approach for common vole population dynamics using historical regional data through 2024.
- Built statistical and machine learning models in R and Python with spatial and temporal validation; reduced 145 candidate variables to 12 key predictors.
- Achieved $R^2 = 0.75$ in geographical validation and >95% accuracy in forecasting population trend direction.
- Identified demographic cycles ranging from approximately 2 to 7 years.

SELECTED PERSONAL PROJECTS

SignalDesk — Python · FastAPI · Finance · Machine Learning — Research tool prioritizing insider-trading filings, combining SEC Form 4 data, market context and anomaly detection into quantitative BUY/SELL/HOLD/IGNORE signals with a public transparency track record.

Music Stats — Python · Streamlit — Interactive application analyzing personal listening history using music metadata and Last.fm data; deployed with Streamlit.

Earnings Calls NLP Pipeline (WIP) — PyTorch · FinBERT · NLP · Streamlit — NLP pipeline analyzing earnings-call transcripts with FinBERT to detect sentiment, executive uncertainty and gaps between prepared remarks and Q&A.

DataPipe (WIP) — FastAPI · React · D3.js · Static Analysis — Code analysis and visualization tool that turns a zipped Python project into an interactive graph of its data flow, from source to output.

Literary Character Stylometry (WIP) — Python · NLP · LLM · spaCy — NLP/LLM pipeline comparing literary characters across books to find psychological and narrative similarities, using a local LLM and multilingual embeddings.

WatchList (WIP) — Python · JavaScript · Supabase · TMDB API — In-progress social movie-tracking application combining user accounts, external APIs, persistent data and statistical analysis.

Movie Stats — Python · Streamlit — Three-year personal project analyzing ~900 films using a curated/scraped dataset, with interactive statistical analyses and visualizations.

Financial Portfolio Optimization — R · Finance · Markowitz Model · Statistics — Analysis and optimization of a seven-stock portfolio (2020–2025): efficient frontier, robust covariance estimation, and comparison of minimum-variance, maximum-return, maximum-Sharpe-ratio and equal-weighted portfolios.

EDUCATION

BSc in Statistics | University of Valladolid (UVa) | Valladolid, Spain | Sep 2021 - Jul 2025

Thesis grade: 9.8 / 10. Thesis: “Predictive Models for Common Vole Population Fluctuations and Agricultural Crop Risk” — early-warning modeling using time-series analysis.

TECHNICAL SKILLS

Programming: Python · R · SQL · JavaScript · Java · C

Data Science & Statistics: Regression · GLM · LASSO · Elastic Net · Random Forest · Gradient Boosting · SARIMAX · ETS · STL · PCA · Clustering · Bayesian Methods · Feature Engineering · Time Series

Data & Visualization: Pandas · NumPy · Scikit-learn · Plotly · Power BI · Looker · Streamlit

Web & APIs: React · FastAPI · REST APIs · Vite · React Router · HTML/CSS

Databases & Tools: PostgreSQL · DuckDB · Supabase · Git · Excel · SAS · MATLAB · SPSS · AWS · Azure

Other: Web Scraping · APIs · AI Agents · RAG · Data Pipelines · Predictive Modeling

CERTIFICATIONS & TRAINING

Renault Group: Looker S5BD Training

LinkedIn Learning: Agentic AI · LangChain & Hugging Face · OpenAI & LangChain · Program Evaluation for Data Science · Advanced NoSQL · AWS ML Specialty (MLS-C01) Cert Prep

LANGUAGES

Spanish: Native · English: Full Professional Proficiency · Italian: Upper-Intermediate / Advanced